AI-based Design Change Impact Analysis system

Objective:
The objective of this project is to build an AI-based Design Change Impact Analysis system that can take a design change as input and automatically identify impacted systems, estimate cost, validate compliance rules, and provide insights using historical data.

Use Cases:

1. Design change impact analysis across multiple systems
2. Cost estimation based on type and magnitude of change
3. Compliance validation using class rules
4. Reference of historical similar changes for better decision-making

Concept Overview:
The system is based on a layered architecture. It uses a backend service to process inputs, a graph database (Neo4j) to model system relationships, and a frontend UI to display results. The intent extraction is currently handled using a simple logic-based approach for the PoC.

Process & Workflow:

1. User enters a design change through the UI
2. Backend processes the input and identifies the affected system
3. Neo4j graph is queried to find impacted systems through relationships
4. Cost model generates estimated values (P10, P50, P90)
5. Class rules and historical changes are retrieved
6. Final response is displayed on the UI

Work Completed:

1. Reviewed and understood the complete PoC architecture
2. Set up Neo4j and created system nodes, relationships, rules, and historical data
3. Developed backend using FastAPI and implemented API endpoints
4. Built frontend UI using React and Tailwind
5. Integrated frontend with backend for end-to-end functionality
6. Added cost estimation, class rules, and historical data handling
7. Resolved API communication issues (including CORS)
8. Tested complete system with multiple scenarios

Advantages:

- Automates impact analysis and reduces manual effort
- Provides faster and structured cost estimation
- Uses graph-based relationships for better accuracy
- Supports decision-making using historical data

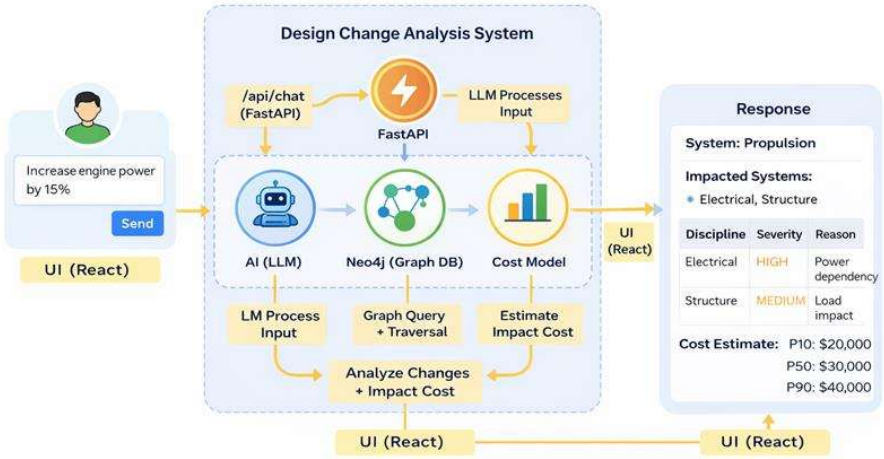
Limitations:

1. Currently uses simplified logic for intent extraction
2. Depends on sample/mock data for PoC
3. Accuracy can be improved with real data and advanced models

Next Steps:

1. Perform detailed testing for different scenarios
2. Improve UI with better visualization and charts
3. Enhance backend logic for more accurate responses
4. Prepare demo and documentation for presentation

Work Flow Diagram



Input Diagram

DCIA — Design Change Intelligence Assistant

PoC · MDL Shipbuilding · Source: PS-1 & PS-2

Increase main engine power by 15% for improved speed

Analyse

Increase main engine power by 15% for improved speed

Replace the shaft generator with a new 3MW unit

Add structural stiffening at frames 40–48 to support a new weapons mount

Increase fuel tank capacity by 20% for extended petrol range

Output

Impact assessment

Design Change Impact Assessment:

User Request:
Increase main engine power by 15% for improved speed

Key Findings:
- Total impacted systems: 6
- Class rules triggered: 4

Cost Estimate:
- P10: $4947188
- P50: $6596250
- P90: $8904938

Analysis:
The proposed change has cascading impacts across multiple ship systems. High severity dependencies should be reviewed carefully.

Conclusion:
The design change is feasible with moderate cost and manageable risk.

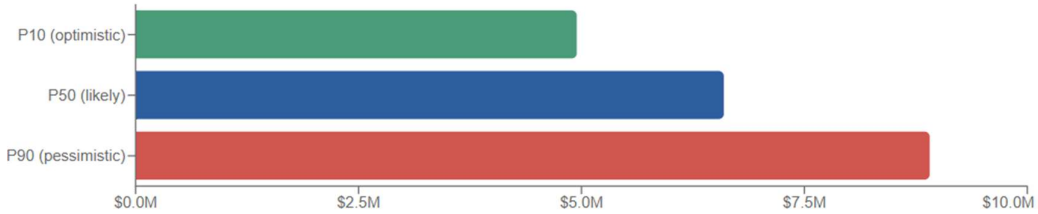
(Note: This is a mock AI-generated report for PoC demonstration.)

Source: Neo4j graph traversal + LLM synthesis · Confidence: MEDIUM

Affected systems (6)

System	Discipline	Hops from origin	Severity	Reason
Hull Structure	Structure	1	HIGH	—
Electrical Power	Electrical	1	MEDIUM	—
Shafting & Gearbox	Propulsion	1	HIGH	—
Fuel System	Outfitting	1	MEDIUM	—
Outfitting / HVAC	Outfitting	2	LOW	—
Weapons Systems	Weapons	2	LOW	—

Cost estimate (USD)



PoC parametric model — replace with MDL historical data regression in production

Class rule flags (4)

LR — Pt3 Ch2 Sec5.3

Main engine power-to-weight ratio limits for naval vessels

Triggered by: Main Propulsion

DNV — RU-SHIP Pt4 Ch2

Vibration limits for hull structure linked to propulsion

Triggered by: Hull Structure

DNV — RU-SHIP Pt4 Ch6

Electrical load balance requirements after machinery change

Triggered by: Electrical Power

LR — Pt3 Ch4 Sec2.1

Shaft alignment tolerances post power upgrade

Triggered by: Shafting & Gearbox

Historical analogues

HC-2022-014 — Engine upgrade from MTU 3600 to MTU 4000 on OPV class

Power increase 12%. Required shaft re-alignment, electrical load re-calc, hull stiffening at frames 42-46.

Actual cost: $3,200,000 · Schedule delay: 28 days

Neo4j Graph

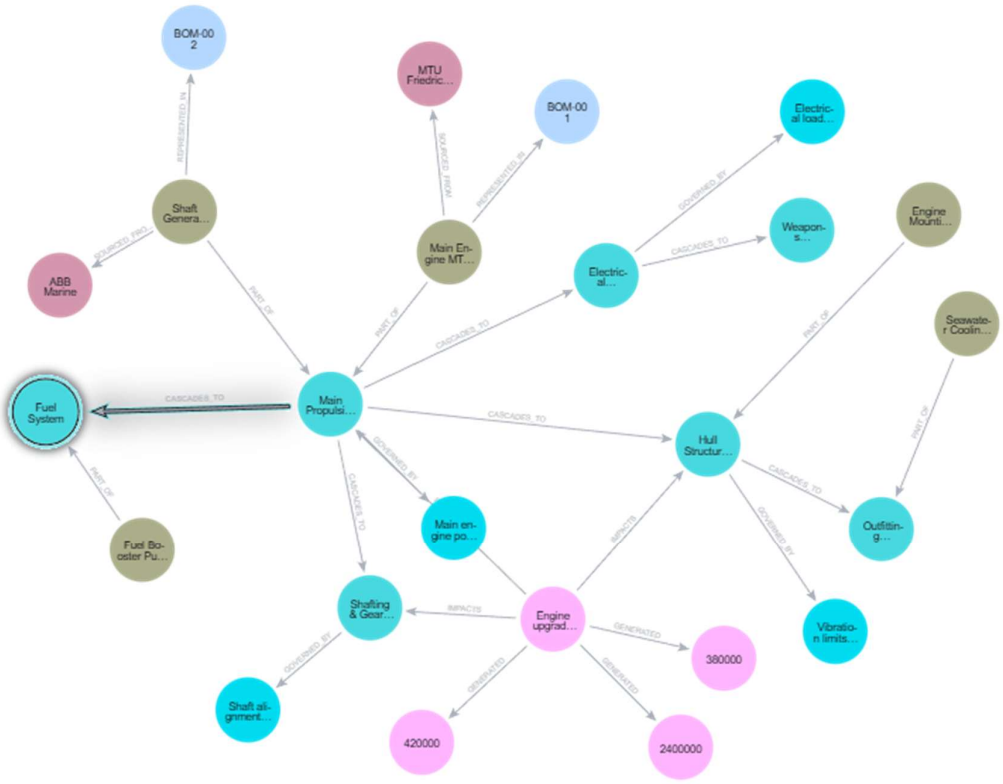

BOM-00 2
MTU Friedric...
BOM-00 1
Electric-al load...
REPRESENTED_IN
SOURCED_FROM
REPRESENTED_IN
GOVERNED_BY
Shaft Genera...
Main En-gine MT...
Electric-al...
Weapon-s...
Engine Mounti...
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ABB Marine
PART_OF
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Main Propulsi...
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Hull Structur...
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Shafting & Gear...
IMPACTS
Engine upgrad...
GENERATED
380000
Vibratio-n limits...
GOVERNED_BY
Shaft ali-gnment...
GENERATED
GENERATED
420000
2400000